ORDER MANAGEMENT SYSTEM

Database Design, Sample Data, Order History & Sales Analysis

MySQL Project Documentation

1. Project Overview

The Order Management System stores customer details, product information, orders, and the individual products included in each order. Primary keys uniquely identify records, while foreign keys maintain relationships between customers, orders, products, and order details. Analysis queries produce customer order history, product demand, customer spending, average order value, and total sales.

2. Orders & Order_Details Table Design

Orders

order_id INT PRIMARY KEY AUTO_INCREMENT; customer_id INT NOT NULL; order_date DATE NOT NULL; total_amount DECIMAL(10,2) NOT NULL DEFAULT 0 CHECK (total_amount >= 0); order_status VARCHAR(20) NOT NULL DEFAULT 'pending' with allowed values pending, shipped, delivered, cancelled; customer_id references customer(customer_id).

Order_Details

order_detail_id INT PRIMARY KEY AUTO_INCREMENT; order_id INT NOT NULL; product_id INT NOT NULL; quantity INT NOT NULL CHECK (quantity > 0); price DECIMAL(10,2) NOT NULL CHECK (price > 0); UNIQUE(order_id, product_id); order_id references orders(order_id) ON DELETE CASCADE; product_id references product(product_id).

3. Updated ER Diagram

Relationship summary:

Parent table	Relationship	Child table
Customer (customer_id)	1 to many	Orders (customer_id FK)
Orders (order_id)	1 to many	Order_Details (order_id FK)
Product (product_id)	1 to many	Order_Details (product_id FK)

Text ER diagram: CUSTOMER 1 ——< ORDERS 1 ——< ORDER_DETAILS >—— 1 PRODUCT. A customer may place many orders. Each order may contain multiple products through Order_Details. A product may be included in multiple orders.

4. Sample Customer, Product, Order & Order_Details Data

Customer

ID	Customer	Email
1	Suresh	suresh@gmail.com
2	Meena	meena@gmail.com
3	Vijay	vijay@gmail.com
4	Anitha	anitha@gmail.com
5	Rahul	rahul@gmail.com

Product

ID	Product	Price (₹)
1	Smartphone	25,000.00
2	Tablet	18,000.00
3	Webcam	2,500.00
4	Laptop Stand	1,500.00
5	Speakers	3,500.00
6	External SSD	6,000.00

Orders

Order ID	Customer ID	Date	Total (₹)	Status
1	1	2026-10-01	28,500.00	delivered
2	2	2026-10-03	20,500.00	shipped
3	3	2026-10-05	9,500.00	pending
4	1	2026-10-07	6,000.00	delivered
5	4	2026-10-10	25,000.00	cancelled
6	5	2026-10-12	38,500.00	shipped
7	2	2026-10-15	3,500.00	delivered
8	3	2026-10-18	18,000.00	pending

Note: The screenshots supplied by the user show sample totals that are not all consistent with the line-item prices. The SQL script includes sample order totals intended to be aligned with the line items where practical; update totals if your submitted database must exactly match existing screenshots.

5. Customer Order History & Purchase Analysis Report

Customer Order History

The order history query joins Customer and Orders. It displays customer ID, customer name, order ID, order date, total amount, and order status. This helps track each customer's order timeline.

Product Purchase Analysis

The product analysis queries show which products appear in order details, how often they are ordered, and the total quantity sold. Cancelled orders are excluded from sales-oriented calculations.

Customer Spending Analysis

The customer spending query sums the total amount for non-cancelled orders and sorts customers from highest to lowest spending. Customers without qualifying orders are still included with a total of zero.

Sales Metrics

Average order value is calculated with AVG(total_amount), and total sales are calculated with SUM(total_amount), excluding cancelled orders. The figures depend on the order totals stored in the Orders table.

6. Query Output Screenshots

The following screenshots were supplied as evidence of query execution in MySQL Workbench.

Screenshot 1: Orders table output (sample order records)

order_id	customer_id	order_date	total_amount	order_status
1	1	2026-10-01	28500.00	delivered
2	2	2026-10-03	21500.00	shipped
3	3	2026-10-05	9500.00	pending
4	1	2026-10-07	6000.00	delivered
5	4	2026-10-10	25000.00	cancelled
6	5	2026-10-12	38500.00	shipped
7	2	2026-10-15	3500.00	delivered
8	3	2026-10-18	18000.00	pending
NULL	NULL	NULL	NULL	NULL

Screenshot 2: Orders table output (verification)

order_id	customer_id	order_date	total_amount	order_status
1	1	2026-10-01	28500.00	delivered
2	2	2026-10-03	21500.00	shipped
3	3	2026-10-05	9500.00	pending
4	1	2026-10-07	6000.00	delivered
5	4	2026-10-10	25000.00	cancelled
6	5	2026-10-12	38500.00	shipped
7	2	2026-10-15	3500.00	delivered
8	3	2026-10-18	18000.00	pending
NULL	NULL	NULL	NULL	NULL

Screenshot 3: Product table output

product_id	product_name	price
1	Smartphone	25000.00
2	Tablet	18000.00
3	Webcam	2500.00
4	Laptop Stand	1500.00
5	Speakers	3500.00
6	External SSD	6000.00
NULL	NULL	NULL

Screenshot 4: Average order value output

average_order_value
17928.57

Screenshot 5: Customer spending analysis

customer_id	customer_name	total_spending
5	Rahul	38500.00
1	Suresh	34500.00
3	Vijay	27500.00
2	Meena	25000.00
4	Anitha	0.00

Screenshot 6: Customers with the highest number of orders

customer_id	customer_name	total_orders
1	Suresh	2
2	Meena	2
3	Vijay	2

Screenshot 7: Total quantity sold by product

product_name	total_quantity_sold
Smartphone	2
Tablet	3
Webcam	2
Laptop Stand	3
Speakers	3
External SSD	2

Screenshot 8: Product order-count analysis

product_id	product_name	times_ordered
1	Smartphone	2
2	Tablet	3
3	Webcam	2
4	Laptop Stand	3
5	Speakers	3
6	External SSD	2

Screenshot 9: Product order-count analysis (alternate output)

product_id	product_name	times_ordered
1	Smartphone	3
2	Tablet	3
3	Webcam	2
4	Laptop Stand	3
5	Speakers	3
6	External SSD	2

Screenshot 10: Order line-item details with customer and product information

product_id	product_name	order_id	customer_name	quantity	price	item_total
6	External SSD	4	Suresh	1	6000.00	6000.00
6	External SSD	3	Vijay	1	6000.00	6000.00
4	Laptop Stand	1	Suresh	1	1000.00	1000.00
4	Laptop Stand	2	Meena	1	1000.00	1000.00
4	Laptop Stand	6	Rahul	1	1500.00	1500.00
1	Smartphone	1	Suresh	1	25000.00	25000.00
1	Smartphone	6	Rahul	1	25000.00	25000.00
5	Speakers	7	Meena	1	3500.00	3500.00
5	Speakers	3	Vijay	1	3500.00	3500.00
5	Speakers	6	Rahul	1	2000.00	2000.00
2	Tablet	2	Meena	1	18000.00	18000.00
2	Tablet	8	Vijay	1	18000.00	18000.00
2	Tablet	6	Rahul	1	10000.00	10000.00
3	Webcam	1	Suresh	1	2500.00	2500.00
3	Webcam	2	Meena	1	2500.00	2500.00

Screenshot 11: Customer order history report

customer_id	customer_name	order_id	order_date	total_amount	order_status
1	Suresh	1	2026-10-01	28500.00	delivered
1	Suresh	4	2026-10-07	6000.00	delivered
2	Meena	2	2026-10-03	21500.00	shipped
2	Meena	7	2026-10-15	3500.00	delivered
3	Vijay	3	2026-10-05	9500.00	pending
3	Vijay	8	2026-10-18	18000.00	pending
4	Anitha	5	2026-10-10	25000.00	cancelled
5	Rahul	6	2026-10-12	38500.00	shipped

7. CRUD Operations

Create: INSERT a new customer, product, or order. Read: SELECT records and run reports. Update: UPDATE an order status or customer information. Delete: DELETE an order; associated order_details rows are removed automatically because of ON DELETE CASCADE.

8. Conclusion

This database provides a structured foundation for managing customers, products, orders, and order line items. Constraints protect data quality, foreign keys preserve relationships, and analysis queries provide useful insights into customer purchases and sales.